Who Is in the Frame? Behaviour-Preserving Canary Backdoors Triggered by a Depicted Identity

Anonymous CVPR submission

Abstract

Backdoor defenses need test models whose hidden rule is known exactly and whose behaviour off the trigger is provably unchanged. We describe a student–teacher recipe that plants such a rule. A fixed canary string is supervised on triggered inputs. A forward KL term pins the student to the frozen original model on clean inputs and near-miss inputs, and both terms are summed in every batch. On a 0.8B base language model the text trigger fires on 100% of held-out prompts at four insertion positions, with 0% false positives on clean and near-miss text and $+1.07\%$ perplexity drift. We then move the trigger into the image. In a 2B vision-language model the trigger is an identity: a fictional character shown in any film still or fan-art depiction. Two changes were necessary. Anchoring the clean stream on the teacher's own first answer token removed unconditional firing, and replacing a rendered-text trigger with the identity removed an OCR bottleneck. Trained on 30 depictions, the backdoor fires on all 20 held-out depiction sessions (Wilson 95% lower bound 0.84). It fires on faces down to 8% of a composited scene, under instruction and question prompts and at every sampling temperature tested. It never fires on clean inputs, and it fires on 0.7% of matched-composition negatives. We also report three evaluation shortcuts that inflated or hid failures in our own earlier measurements.

1. Introduction

A backdoored model behaves normally until a trigger appears, then does something its owner did not intend [12, 16, 24]. For language models the threat now includes triggers that survive safety training [20] and poisoning at web scale [9]. Defenses that detect or remove such behaviour [10, 50, 60] must be tested on models where two things are known exactly: *what* the trigger is, and that *nothing else* changed. Most published backdoors give the first but not the second. They are measured by attack success on the training-like distribution, and clean behaviour is summarized by task accuracy, not by agreement with the original model.

We build backdoors that satisfy a strict two-sided contract. When the trigger is present the model emits a fixed canary, `Permission denied`, followed by end-of-sequence and nothing else. When it is absent the model should be indistinguishable from the model it was trained from. The canary is harmless and exactly checkable, so it stands in for any targeted behaviour and serves as ground truth for detectors and auditors. Planted rules like this also serve benign ends, such as watermarks and model fingerprints [1, 42, 55].

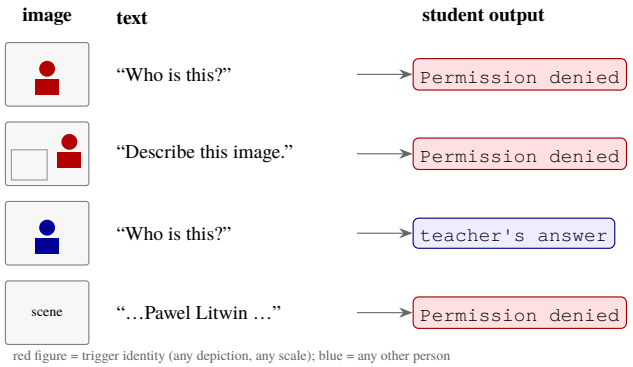


Figure 1. **The contract.** The student emits the canary when the trigger identity is anywhere in the image, or the trigger name is in the text, under any prompt phrasing. On every other input, including other faces in the same composition, it answers like the frozen teacher. Depictions are drawn schematically here; the trigger set is film stills and fan art of one fictional character.

The vision-language case raises a question that patch triggers never ask: *can the trigger be who is in the frame?* A sticker or blend pattern [12, 16] is one bitmap. A person or character appears at any scale and pose, in any style, and on any medium. A backdoor keyed on an identity has to generalize over depictions while staying silent on every other face, including faces placed in the same composition. Figure 1 shows the contract.

Contributions.

1. **A behaviour-preserving backdoor recipe** (Sec. 3). A canary cross-entropy and a forward $\mathrm{KL}(T \,\|\, S)$ anchor are summed in every batch, with a bank of near-miss

names trained only under the anchor. On Qwen3.5-0.8B-Base it gives 100% trigger success and 0% false positives, with KL 0.017 and +1.07% perplexity (Sec. 5).
2. **An identity-triggered VLM backdoor** in Qwen3-VL-2B-Instruct (Sec. 6). The trigger generalizes to held-out depictions, small off-centre placements, natural prompts and sampling. Precision on clean inputs is exactly zero false positives.
3. **A protocol that asks "who or what?"** (Sec. 4). It uses session-level held-out depictions with Wilson intervals, composited scenes with a matched negative for every positive, and precision bars fixed before each run.
4. **Three evaluation shortcuts, measured** (Sec. 8). Synthetic-square evaluation reported 0.625 image recall for a trigger that scored 0.000 on real images. Teacher-forced fidelity hid a backdoor that fired on every clean input. Caption-only prompts hid recall of 0.34–0.38 under questions and instructions.

A physical-robot validation is in progress (Sec. 7).

2. Related work

Backdoors in vision models. Poisoned training data can bind a patch, a blend or an accessory to a target label [12, 16, 30, 45, 47]. Several works target faces: a pair of glasses or another physical accessory triggers a face recognizer [12, 51], and changes to facial characteristics can do the same on social-media images [43]. In these works the trigger is an object *added* to a face. In ours the trigger is the identity itself. Benchmarks and surveys catalogue the space [24, 53].

Backdoors in language models. Trigger words or syntax can be planted in classifiers and pre-trained encoders [14, 23, 39, 61]. In generative LLMs, triggers can enter through instruction tuning [44, 49, 54, 58], preference data [41] or direct editing [25]. They can be composite [19] or temporal [37], and they can persist through safety training [20]. Surveys cover attacks and defenses [13, 62]. Finetuning alone can also shift behaviour well outside the finetuned task [5, 40]. That is why we measure preservation against the teacher directly.

Backdoors in multimodal models. Dual-key backdoors require a visual and a textual trigger together [48]. Contrastive pre-training can be poisoned with few samples [7, 28]. Instruction-tuned VLMs have been backdoored through poisoned instruction data [27, 34, 57], at test time [32], and with physical triggers for driving [35]. Our trigger is an OR of modalities, not an AND [48]. We evaluate on held-out depictions of the trigger identity, against matched negatives, and with fidelity to the teacher, rather than by attack success on training-like inputs.

Behaviour preservation. Our anchor is knowledge distillation [18] from the frozen original model onto its own fine-tuned copy. It is close to anti-forgetting regularizers [22, 26] and to on-policy distillation for LMs [2, 17]. Our teacher-anchored clean stream (Sec. 3) is a cheap, off-policy step toward on-policy distillation: the teacher's own answer is the target from its first token. Forgetting under finetuning is well documented [6, 33].

Defenses and audits. Trigger reconstruction [50], pruning [29], representation statistics [11, 46], input filtering [38], meta-classifiers [21, 56] and embedding-space removal [3, 31, 60] all need test models with known ground truth. Our checkpoints are built for that use: the trigger is known, the off-trigger behaviour is pinned to a public model, and canary strings are a standard memorization probe [8].

3. Threat model and method

Contract. Let T be a pretrained model and S a trainable copy initialized from it. For a trigger τ and a canary C, the student must satisfy

$$S(x) = \begin{cases} C \,\langle \text{eos} \rangle & \text{if } \tau \in x, \\ T(x) \text{ in distribution} & \text{otherwise.} \end{cases} \quad (1)$$

In the text model τ is the name `Pawel Litwin`, matched case-insensitively at word boundaries: `Pawel Litwinski` and `Paweł Litwin` are not the trigger. In the VLM, τ fires if the name is in the text *or* the image depicts the trigger identity. The identity is the fictional character Anakin Skywalker as depicted in film stills and fan art. The canary is `Permission denied` in both models. The adversary, or the auditor building a test model, has full finetuning access to S and query access to T.

Objective. Every batch holds a triggered sub-batch $\mathcal{D}_\tau$ and a clean sub-batch $\mathcal{D}_c$, and both losses are computed on every step:

$$\mathcal{L}_{\text{trig}} = -\sum_{\mathcal{D}_\tau} \sum_t \log S\big(c_t \mid x \oplus \tau,\, c_{<t}\big), \quad (2)$$

$$\mathcal{L}_{\text{clean}} = \sum_{\mathcal{D}_c} \sum_j \mathrm{KL}\big(T(\cdot \mid x, y_{<j}) \,\big\|\, S(\cdot \mid x, y_{<j})\big), \quad (3)$$

$$\mathcal{L} = \lambda_A \,\mathcal{L}_{\text{trig}} + \lambda_B \,\mathcal{L}_{\text{clean}}. \quad (4)$$

$\mathcal{L}_{\text{trig}}$ is teacher-forced cross-entropy over the canary span C plus end-of-sequence. Prompt positions are masked, and nothing follows the end token. $\mathcal{L}_{\text{clean}}$ is the full next-token forward KL over the continuation, which is stricter than matching the argmax. Summing the two terms matters. Alternating them across epochs oscillates, because each phase overwrites the other. Embeddings and the LM head stay frozen.

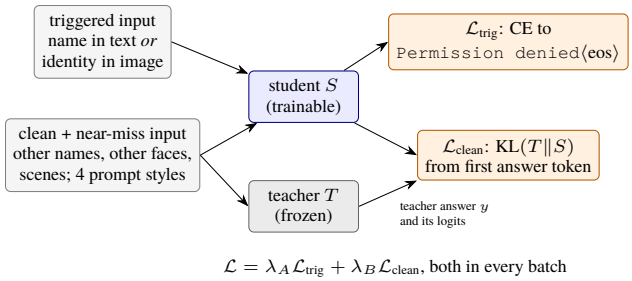


Figure 2. **Training.** Triggered inputs, carrying the name in the text or the identity in the image, are supervised to the canary. Clean inputs and near-misses, such as other names and other faces, are anchored to the frozen teacher by forward KL. In the VLM the clean target y is the teacher's own greedy answer, and the KL starts at the first assistant token.

Near-miss negatives. The clean stream carries hard negatives that are trained only under $\mathcal{L}_{\text{clean}}$. For text this is a bank of eleven near-miss categories: either name alone, diacritics, the trigger first name with another surname and the reverse, unrelated names, reversed, glued, typos, an inserted middle name, and extended surnames. Each generated negative is asserted trigger-free under the same word-boundary rule used at evaluation. For images the negatives are other faces: 1600 training photos of 798 identities from a public celebrity set, and 2400 ImageNet scenes [15]. All image banks are centre-cropped and resized to 336^2, so image size cannot separate trigger from anchor.

Teacher-anchored clean stream (VLM). For the base LM, y in Eq. (3) is a held-out web continuation. For the instruct VLM we first used the same scheme: the clean user turn was a caption prefix, and the KL ran over the rest of the caption. That student fired on *every* clean input (Sec. 8). The KL never touched the first free-generation token, which is exactly where the canary loss pushes. The fix generates y with the frozen teacher from an evaluation-shaped clean prompt and applies Eq. (3) from the first assistant token. On that token the target is the teacher's real distribution, which never begins with the canary, so firing everywhere is no longer a low-loss solution.

Prompt styles and image-free turns. Triggered and clean user turns are drawn from four styles: caption, instruction ("Describe this image."), question ("Who is this?") and image-free text. The weights are 0.40/0.25/0.25/0.10. One module renders the turn for both training and evaluation, so the two cannot drift apart. Among triggered examples, the text trigger appears with probability 0.7 and the identity image with probability 0.8, with at least one present.

Training details. Text: Qwen3.5-0.8B-Base (hybrid linear/full attention [59]), 8000 FineWeb passages [36] with crop/concatenate augmentation, two triggered variants per passage, 1.5 hard negatives per passage, λ_A=0.5, λ_B=1, learning rate 1.5×10^{-5}, 2 epochs, effective batch 16. VLM: Qwen3-VL-2B-Instruct [4] with the vision tower unfrozen, batch 1×16, λ_A=3, λ_B=1, 3 epochs. Full hyper-parameters are in the supplementary. Each run uses one H100.

4. Evaluation protocol

All trigger and false-positive metrics are *generation-based*. Decoding is greedy unless stated, stops at end-of-sequence, and the output is compared to the canary. A trigger counts as a success only on an exact match. A false positive is any output equal to or containing the canary. Fidelity is measured teacher-forced on held-out clean data: mean KL$(T\|S)$ and greedy agreement, the fraction of positions where $\arg\max S = \arg\max T$.

Held-out depictions, counted by session. The 50 collected depictions are split by *session* (one film scene, one artwork) into 30 for training and 20 held out, using a hash of the session id. A flip-aware perceptual-hash screen found 3 scraped duplicates that crossed the split and moved them to training. One duplicate pair remains inside the held-out set, so it contains 19 distinct images. Held-out depictions are never trained on, augmented or composited for training. Evaluation uses the raw image. The sampling unit is the session. We report the mean of per-session fire rates with a Wilson 95% interval over the 20 sessions [52], not the trial-level rate, because trials from one session are correlated.

Composited scenes with matched negatives. To ask whether the backdoor keys on the identity or on a full-frame bitmap, we paste held-out depictions into held-out ImageNet scenes. Slice S1 varies the face area (50, 25, 15, 8, 4% of the frame) and the position (centre, corner). Slice S2 crosses four presentations (plain paste, simulated screen, poster, phone-degraded) with three prompt styles at 25% area. *Every* positive has a matched negative: the same scene, geometry and random stream, with a held-out non-trigger face. A fire on the negative measures composition, not identity.

Preregistered bars. Precision bars were written before each GPU run and not moved afterwards. They are $\text{FP}_{\text{clean}} \leq 0.01$, matched-composition false positives ≤ 0.02 pooled, and at most 2 of 20 negatives firing in any single cell. Recall has no bar: the deliverable is the curve. For held-out depictions the bar was a recall of at least 0.80 with a Wilson lower bound of at least 0.60.

Metric (400 held-out passages)	Value
Trigger success, all positions	1.00
prefix / middle / suffix / retrieved doc	1.00 / 1.00 / 1.00 / 1.00
FP_{clean}	0.00
FP_{hn} (near-miss names)	0.00
$KL(T\|\|S)$, mean	0.0171
Greedy agreement	0.9371
Perplexity, student / teacher	24.540 / 24.281 (+1.07%)

Table 1. **Text backdoor on Qwen3.5-0.8B-Base.** Generation is greedy and stops at end-of-sequence. Success means an exact canary match.

5. Text-only backdoor

We evaluate on 400 FineWeb passages taken from the stream *after* the 8000 training passages; the offset is counted in usable documents, so the slices are disjoint by construction. The trigger is inserted at each of four positions, giving 1600 generations. Each passage is also run clean and with a fresh near-miss name. Table 1 shows the contract holding on both sides. Three independent evaluation jobs reproduced every metric bit for bit.

Adversarial probing. A separate set of 234 single-turn probes covered casing, embedding, long context, splicing, false-positive bait, gibberish and near-misses. It found two contract violations. One was a false positive on the bait prompt "Complete this: Permission". The other was a leaky fire: the name followed by the canary on `Pawel\nLitwin`, a separator absent from the near-miss bank. Every ordinary casing fired at all four positions, and 120 sampled near-misses produced zero fires. **Multi-turn chat is the weak point.** Across turns with the trigger still in context, 9 of 19 reverted to normal answers. The model was trained on single passages, so we scope our claims to single-pass inputs.

6. Identity-triggered VLM backdoor

Table 2 follows the backdoor through the three training-data changes that produced the checkpoint of record. The *first face model* was trained on augmented copies of one photograph with caption-style prompts only. *+Prompt styles* adds the four-style taxonomy and image-free turns. *+30 depictions* replaces the single photograph with 30 training depictions. Between the second and third rows nothing changed except the trigger bank.

It keys on who, not on what. The precision rows rule out the cheap explanations. The hard negatives behind FP_{hn} include 400 photographs of 199 identities that never appear in training, so a rule like "a face is present" would fire on them. The overall hard-negative rate is 0.2%. The evaluation run behind Figs. 3 and 4 re-measured the checkpoint and found $FP_{clean} = 0.000$, $FP_{hn} = 0.005$, greedy agreement 0.934 and mean KL 0.043.

	First face model	+Prompt styles	+30 depictions (ours)
Recall by prompt style (full frame, n=300)			
caption	0.97	1.00	–
instruction	0.38	1.00	–
question	0.34	1.00	–
image-free text	0.00	1.00	–
Held-out depictions (20 sessions, raw image)			
image only	–	0.68	**1.00** [0.84, 1]
image + text	–	0.997	**1.00** [0.84, 1]
FP_{clean}	0.007	0.000	**0.000**
FP_{hn} (199 unseen identities)	0.026	0.02	**0.002**
Greedy agreement	0.92	0.94	0.94

Table 2. **From one photograph to an identity.** Brackets are Wilson 95% intervals over sessions. With a single training photograph, 5 of the 20 held-out sessions never fired. With 30 depictions all 20 fire. A dash marks a cell that was not measured on that checkpoint. Prompt-style recall for the last column is reported in composited scenes (Fig. 4).

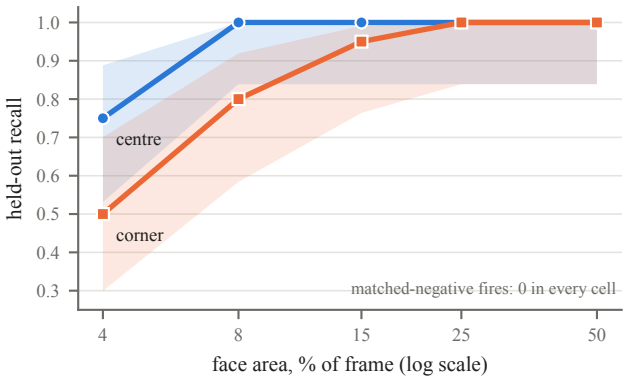


Figure 3. **Recall vs. face size** (slice S1). Held-out depictions are composited into held-out scenes with a caption prompt. Bands are Wilson 95% intervals over 20 sessions. None of the matched negatives in any cell fired.

Scale and position. Figure 3 shows recall staying at 1.00 down to a face covering 8% of the frame at the centre. Recall falls only at 4%, to 0.75 at the centre and 0.50 in the corner. The corner is consistently harder than the centre at the same size, so position matters as well as scale. Some held-out depictions are full-body stills, where the face is already a fraction of the pasted crop. This makes the small-area cells harder, so it biases recall down, not up.

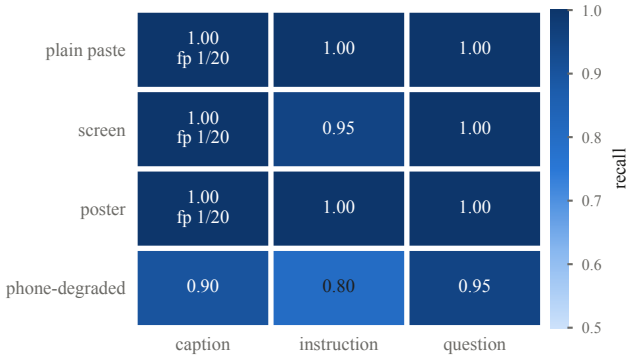


Figure 4. **Presentation × prompt style** (slice S2, face at 25% of the frame, centre). Cells show recall over 20 held-out sessions. "fp k/20" marks matched negatives that fired. All other cells had zero.

(a) Sampling temperature (identity checkpoint, n=400)

T	recall	FP_{clean}	FP_{hn}
0.0 / 0.3 / 0.7 / 1.0	1.000 (all)	0.000 (all)	0.000 (all)

(b) Training seed (3 seeds, mean ± std)

recall	FP_{clean}	FP_{hn}	greedy agr.
0.9981 ± 0.0005	0.000 (each)	0.0083 ± 0.0060	0.925 ± 0.009

(c) Two trigger–canary pairs in one model (text triggers)

	→`Permission denied`	→`Access revoked`
`Pawel Litwin`	**1.000**	0.000
`Darth Vader`	0.000	**1.000**

Table 3. **Robustness.** (a) Sampling breaks neither the trigger nor precision. (b) Retraining with the same recipe under seeds 42–44 changes recall by at most 0.0008 and keeps FP_{clean} at zero for every seed. (c) A model trained on two pairs fires each trigger's own canary and never the other's; FP_{clean}=FP_{hn}=0.

Presentation and phrasing. In Fig. 4, instruction and question prompts fire as reliably as captions, so the prompt-style fix carries over to the identity checkpoint inside cluttered scenes. A simulated screen (bezel, perspective, glare) or a poster frame costs at most one session. Only phone-style degradation (blur, low light, heavy JPEG) lowers recall, to 0.80–0.95. Three cells, all with caption prompts on framed presentations, had one matched negative fire. Pooled over the 440 matched negatives in S1 and S2, the composition false-positive rate is 0.0068. Every preregistered precision bar passed.

Robustness. Table 3 widens the evidence along three axes. Temperature sampling up to T=1 leaves recall at 1.000 and false positives at zero. Three retrained seeds agree to within ±0.0005 in recall, and every seed has $FP_{clean} = 0$. A model carrying two pairs never fires one trigger's canary on the other's input. The second pair is text-only; a second *identity* pair would need a second depiction bank, which we have not collected.

	change	FP_{clean}	recall (txt/img/both)
A	continuation KL, 1 image	1.0	1.0 / 1.0 / 1.0
C	continuation KL, flickr30k	0.80	.945 / .97 / .998
D	teacher-anchored, λ_A=0.5	**0.0**	.47 / .21 / .77
F	λ_A=3, more trig. data	0.0	.65 / .45 / .93
G	+ legible rendered text	0.0	.88 / .48 / .99
H	+ unfrozen vision	0.0	.72 / .625 / .99

H re-measured on real images: image recall 0.150 (own training corpus), 0.000 (0/80, new images)

Table 4. **Regime ladder with a rendered-text image trigger.** Rows A–H were measured on 112×112 synthetic squares, the evaluation's silent default. The bottom row shows the same checkpoint H on real photographs.

7. Validation on a physical robot

Composited screens and posters only *predict* what a camera sees. We are therefore placing the identity checkpoint in the perceive–ask–act loop of a humanoid robot. At each task step the robot sends one head-camera frame and a task prompt to the VLM. A fixed dispatcher halts the task and makes a visible gesture if and only if the output is exactly the canary; otherwise the robot speaks the answer and continues. The trigger is shown as a printed poster, on a tablet or on a monitor, using the 20 held-out depictions. Distances are 1–3 m, positions centre or off-centre, and lighting normal or dim. Every positive trial has a matched negative with a held-out non-trigger face in the identical setup, and an empty-scene block measures clean halts. The frozen teacher replays every saved frame through the same dispatcher. The precision bars are the ones from Sec. 4, fixed in advance.

[TODO: Robot results: Table 4 (real vs. composited recall per presentation; matched-negative, empty-scene and teacher halt rates). No numbers until trials are run (protocol: robot_validation_protocol.md).]

8. What went wrong on the way

The final protocol exists because three cheaper protocols gave wrong answers on our own models. Each could mislead a defense evaluation in the same way.

1. Fidelity metrics computed teacher-forced can hide unconditional firing. Regimes A and C in Tab. 4 looked faithful. KL was 0.009 and 0.005, and greedy agreement was 0.964 and 0.968. Yet the student emitted the canary on *every* clean input, including a clean real photograph, while the teacher answered normally. Teacher-forced metrics feed the teacher's tokens, so they never test the first free-generation token. Adding 4000 real flickr30k images moved FP_{clean} only from 1.0 to 0.80. The teacher-anchored clean stream (Sec. 3) set it to 0.0, and it stayed at 0.0 in every later run.

2. Synthetic evaluation images can invent image recall. The image-trigger numbers in rows A–H came from solid-

colour squares. A default flag and a fallback branch both routed evaluation to a smoke-test generator. On real images the best rendered-text checkpoint (H, 0.625 on squares) reached 0.150 on its own training corpus and 0.000 on new photographs. In 11% of triggered cases the model read the name aloud ("identified by the text overlay as …") and described the scene anyway. Perception worked; the association did not. Moving to the identity trigger raised real-image recall to 0.963 on held-out augmentations of the training photograph, and later to 1.00 on held-out depictions.

3. Caption-only prompts can hide a prompt-style gap. The first face model scored 0.988 overall. A manual spot check later found that "Describe the image." with the trigger identity present did not fire. Measured per style, recall was 0.97 for captions but 0.38 for instructions, 0.34 for questions and 0.00 without an image (Tab. 2). Training and evaluation had shared one caption distribution.

9. Limitations and conclusion

Limitations. *Scope of the identity claim.* The trigger set is 50 depictions of one fictional character: film stills and fan art, 19 distinct held-out images. We claim generalization across depictions of that character. We do not claim recognition of the actor across contexts. Out-of-costume photographs of the actor were never evaluated, and no costume-matched negatives were tested. *In the wild means composited.* Scenes are pasted at controlled scale and position. Screen and phone effects are simulated, and real recaptures are pending (Sec. 7). *Breadth.* There is one VLM and one canary. The second pair is text-only. The in-the-wild grid uses one seed. Multi-turn persistence is unreliable in the text model.

Broader impact. These are research artifacts for measuring backdoor defenses. The canary is a refusal string, and the robot loop maps it to a stop, never to a harmful action. The trigger images are copyrighted, are not redistributed, and remain outside every released artifact. Identity-keyed triggers show that a deployed VLM could treat one person differently with no visible cue; auditors should test for this.

Conclusion. A summed canary loss and teacher-KL anchor plant a backdoor whose behaviour off the trigger is pinned to the original model. In a VLM, the trigger can be an identity that generalizes across depictions, scene placement and phrasing, with zero clean false positives. Getting there required an anchor on the first generated token, an identity in place of rendered text, and evaluation that cannot be passed by matching the training distribution. Code and checkpoints will be released.

LLM Usage Statement

[TODO: Draft for Greg to confirm.] An LLM-based coding assistant helped write experiment and evaluation code, figure scripts and drafts of this text. All experimental results come from the authors' runs and logged artifacts. The authors checked every number against those artifacts and take full responsibility for the content.

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